



Thoshith R R 250701797-2025-CSE ▾

T2

Started on Friday, 21 August 2026, 3:31 PM

State Finished

Completed on Friday, 21 August 2026, 3:44 PM

Time taken 13 mins 5 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int function(int n){
3      int c=0;
4      for(int i=n/2;i<n;i++){
5          c++;
6          for(int j=1;j<n;j=2*j){
7              c++;
8              for(int k=1;k<n;k=k*2){
9                  c++;
10             }
11         }
12     }
13     return 2* c+2;
14 }
15 int main(){
16     int n;
17     scanf("%d",&n);
18     int result=function(n);
19     printf("%d",result);
20     return 0;
21 }
22
23
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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